Question1: MSQ (15) Minutes [ ] / 10 Marks

1	2	3	4	5	6	7	8	9	10
D	C	A	B	E	C	A	C	D	B

1. **Examples of $O(1)$ algorithms are:**

   A. Multiplying two numbers
   B. Assigning some value to a variable
   C. Displaying some integer on console
   D. All of Above

2. **What will be the cost function T(n) of the following code?**

```
int value = 0;
for(int i=0;i<n;i++)        n
    for(int j=0;j<i;j++)    (n-1)
        value += 1;
```

   A. n
   B. $(n+1)$
   C. $\frac{n(n-1)}{2}$
   D. $\frac{n(n+1)}{2}$

1st loop will run for n times and another for loop will be run for (n-1) times as the inner loop will only run till the range i which is less than n. overall n(n-1)

3. **If f(n) = O(g(n)) and g(n) = O(h(n)), then**
   A. h(n) = Ω(f(n))
   B. h(n) = O (f(n))
   C. h(n) = Θ (f(n))
   D. None of the above

4. **If f(n) = 5nlogn and g(n) = n log 5n then**
   A. f = O (g)
   B. f = Θ(g)
   C. f = Ω (g)

5. **Given the following two functions: $\sqrt{n}$ and $n^{\sin(n)}$, where $n > 0$, then:**
   A. $\sqrt{n} = O(n^{\sin(n)})$
   B. $n^{\sin(n)} = O(\sqrt{n})$
   C. $\sqrt{n} = \Omega(n^{\sin(n)})$
   D. $n^{\sin(n)} = \Omega(\sqrt{n})$
   E. All answers are incorrect.

6. **Let $f(n) = 2^{2n}$, then a correct asymptotic upper bound for $f(n)$ is:**
   A. $O(2)$
   B. $O(2^n)$
   C. $O(2^{2n})$
   D. All answers are correct.
   E. All answers are incorrect.

7. **Which of the following approaches is adopted in Divide & Conquer algorithms?**

   A. Top-down
   B. Bottom-up
   C. Both
   D. None of the above.

8. **The solution of this recurrence $T(n) = 8\,T(n/2) + 1000n^2$ is (*Hint: use Master method*):**

   A. $O(n^2)$
   B. $O(n)$
   C. $O(n^3)$
   D. $O(n \log n)$

9. **The worst case running times of Insertion sort and Merge sort, respectively, are:**

   A. $O(n \log n), O(n \log n)$
   B. $O(n \log n), O(n^2)$
   C. $O(n^2), O(n^2)$
   D. $O(n^2), O(n \log n)$

10. **A sorting technique is called stable if:**

    A. it takes $O(n\ log\ n)$ time
    B. Relative order of occurrence of non-distinct elements is maintained
    C. It uses divide and conquer technique
    D. It takes $O(n)$ space

Question 2 **Fundamentals of Algorithms Analysis** **(15) Minutes [ ] / 5 Marks**

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1. ***Given the following pseudocode, compute the running time function $T(n)$ in terms of n. [3 marks]***

```
1   function( n ){
2     for i = 1 to n*n
3       for j = 1 to n
4         print "*";
5   }
```

cost | time
c_1 | 0.5
c_2 | $n*n-0+1 = n^2+1$
c_3 | $n(n^2+1)$ ~~$(n+1) = n^3+n^2+n+1$~~
c_4 | n
0.5 | 0.5

$c_2(n^2+1) + n(n^2+1)c_3 + c_4 n$

Possible solution:

$$T(n) = c_2(n^2+1) + ((c_3(n+1))n^2 + c_4 n$$
$$= c_2 n^2 + c_2 + (c_3 n + c_3)n^2 + c_4 n \quad \text{2.5}$$
$$= c_2 n^2 + c_2 + c_3 n^3 + c_3 n^2 + c_4 n$$
$$= n^2(c_2 + c_3) + c_4 n + c_3 n^3 + c_2 = O(n^3) \quad \text{0.5}$$

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2. Find O, c, n_0 of $f(n)= \log (n+5)$ [2 marks]

$f(n) = \log (n+5) \leq \log (2n) \quad \forall n \geq 5$

$\leq \log 2 + \log n = 1 + \log n$

$\leq \log n + \log n = 2 \log n \quad \forall n \geq 2$

$= O(\log n) \quad c=2 \text{ and } n_0=2$

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Question 3 **Recurrence** **(20) Minutes [ ] / 5 Marks**

1. **Using the recursion tree method to solve the following questions for the recurrence $(n) = 3T(n/4) + n^2$ [4 marks]**

What is the sub-problem size at level i? [0.5 mark]

The size of each sub-problems of level i is: $n/4^i$

What is the total cost over all nodes at level [0.5 mark]

The total cost at level i is: $= (3/16)^i * n^2$

How many levels in the recursion tree? [1 mark]

$\text{Log}_4 n + 1$ levels

Calculate the total cost (*Big-O*) of *T(n)* of all levels. Show all the details [2 marks]

$$T(n) = cn^2 + \frac{3}{16}cn^2 + \left(\frac{3}{16}\right)^2 cn^2 + \cdots + \left(\frac{3}{16}\right)^{\log_4 n-1} cn^2 + \Theta(n^{\log_4 3})$$

$$= \sum_{i=0}^{\log_4 n-1} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3})$$

This looks complicated but we can bound it (from above) by the sum of the infinite series

$$\sum_{i=0}^{\infty} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3}) = \frac{1}{1-(3/16)} cn^2 + \Theta(n^{\log_4 3})$$

Since functions in $\Theta(n^{\log_4 3})$ are also in $O(n^2)$, this whole expression is $O(n^2)$. Therefore, we can guess that $T(n) = O(n^2)$.

$f(n)=n^2$

$a=4$ | $\log_4 4 = 1$

$b=4$

$n < n^2 \Rightarrow$ case 3

$4\left(\frac{n}{4}\right)^2 \leq cn^2$

$\frac{1}{4} \leq c \Rightarrow$ OK

$T(n) = O(n^2)$

2. **Solve using Master method the recurrences [1 marks]**

$4\,T(n/4) + n^2$

big $O(n^2)$

Question 4 **Sorting** **(20) Minutes [ ] / 5 Marks**

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1. Apply **Insertion sort** to sort the list {89, 45, 68, 90, 29, 34, 17}. **[1 mark]**

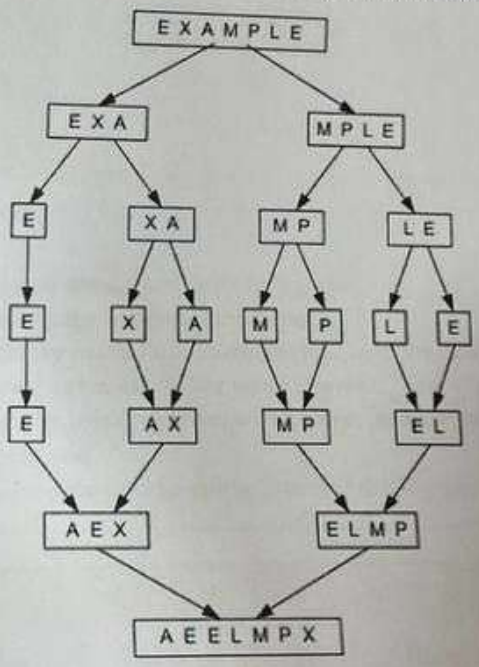
89 |	**45**	68	90	29	34	17
45	89 |	**68**	90	29	34	17
45	68	89 |	**90**	29	34	17
45	68	89	90 |	**29**	34	17
29	45	68	89	90 |	**34**	17
29	34	45	68	89	90 |	**17**
17	29	34	45	68	89	90

I. What is the Time Complexity of **Insertion Sort**? **[0.5 marks]**
Worst case $O(n^2)$,Average case $O(n^2)$
Best case $O(n)$

II. Is **Insertion sort** a stable sorting algorithm? **[0.5 mark]**
Yes, insertion sort is stable algorithm

III. Is **Insertion Sort** an in-place sorting algorithm? **[0.5 mark]**
Yes, insertion sort is an in-place sorting algorithm

2. Apply **Merge sort** to sort the list {*E, X, A, M, P, L, E*} in alphabetical order. **[1 mark]**



I. Is **Merge sort** a stable sorting algorithm? **[0.5 mark]**
Yes, Merge sort is stable

II. Is **Merge sort** an in-place algorithm? **[0.5 mark]**
No, in merge sort the merging step requires extra space to store the elements.

III. What is the Time Complexity of **Merge sort**? **[0.5 mark]**
$O(n\ logn)$